# IMPACT ANALYSIS

## CR-003: Medicine Price Validation and Role-Based Discount Management

### 1. Change Summary

CR-003 introduces Admin-controlled discount options and stronger price validation in the Sales & Billing module. Only the Admin role can add or manage discount options. Pharmacists and Pharmacy Staff can apply an active approved discount only by selecting it from the dropdown on the Sales page. The backend independently retrieves medicine prices, validates the selected discount and calculates the final sale values to protect financial integrity.

### 2. Impact Assessment

| Area | Impact | Assessment |
|---|---|---|
| Sales Frontend | Moderate | Replace free-text discount entry with a dropdown containing active, approved discount options and live calculations. |
| Admin Interface | Moderate | Provide an Admin-only interface to add and manage approved discount options. |
| Backend | High | Validate roles, retrieve database selling prices, whitelist discount options and calculate sale totals. |
| API | Moderate - High | Add discount retrieval and management endpoints, and enforce access restrictions and validation responses. |
| Database | Moderate | Add discount_options and verify or add the price and discount snapshot fields in sale_items. |
| Authentication / Authorization | High | Enforce Admin-only discount management while preserving sales access for all three roles. |
| Sales / Stock | High | Ensure validated calculations and stock changes are committed consistently within the sales transaction. |
| Reporting / Receipts | Moderate | Support reporting of gross revenue, discounts granted and net income; show discount details on receipts. |
| Performance | Low | Discount lookup and monetary calculations add limited processing per sale. |
| Deployment | Moderate | Deploy database, backend and frontend changes together and run role and sales regression tests. |

### 3. Role and Permission Impact

| Role | Allowed Operation | Access Restriction |
|---|---|---|
| Admin | Add and manage discount options in the Admin discount configuration interface; select a discount during sales. | Discount management endpoints are restricted to Admin. |
| Pharmacist | Select an active approved discount from the Sales page dropdown. | Cannot add, edit or deactivate discount options. |
| Pharmacy Staff | Select an active approved discount from the Sales page dropdown. | Cannot add, edit or deactivate discount options. |

The Sales page dropdown is the permitted method for Pharmacists and Pharmacy Staff to apply discounts. Hiding or disabling an Admin control in the browser is not sufficient by itself; the server must reject unauthorized discount-management requests.

### 4. Frontend Impact

- Sales page: replace the free-text discount input with a dropdown populated dynamically from GET /api/sales/discounts.
- The dropdown must show active approved options only and default to 0%.
- Pharmacists and Pharmacy Staff can select an option but must not see or use discount-option management controls.
- The Admin interface provides the controls to add and manage discount options.
- Update line-item Discount Amount, Subtotal, Gross Total, Total Discount and Final Amount when a selection or quantity changes.
- Display currency using the local Rs. format with thousands separators and two decimal places.
- Show clear messages when discount data cannot be loaded or a backend validation error is returned.

## 5. Backend, API and Database Impact

The sales route must retrieve the authoritative medicine selling price from the medicines table. It must validate each submitted discount against the active Admin-configured options and recalculate all monetary values on the server. Client-supplied selling_price, discount_amount and subtotal values must not be trusted.

The Admin discount routes (GET /api/admin/discounts, POST /api/admin/discounts and PUT /api/admin/discounts/:id/status) must enforce authenticateToken and authorizeRoles('admin'). The Sales dropdown retrieval endpoint (GET /api/sales/discounts) and sale creation endpoint (POST /api/sales) remain available to authenticated Admin, Pharmacist and Pharmacy Staff users, subject to existing sales permissions.

The database impact includes a discount_options table with a unique percentage value and active status, plus sale_items snapshot fields for selling price, discount percentage, discount amount and subtotal. The source specification anticipates these schema changes; existing database structure must be reviewed before applying migrations.

## 6. Security Impact

- Enforce Admin-only discount configuration at the backend, not only through frontend visibility or navigation controls.
- Reject Pharmacist and Pharmacy Staff attempts to add, edit or deactivate discount options with 403 Forbidden.
- Do not trust discount percentages or monetary totals supplied by the browser; validate against active database options.
- Retrieve medicine selling prices from the medicines table rather than accepting client-submitted prices.
- Reject unapproved, inactive, negative or out-of-range discount values with a clear client error response.
- Maintain existing authentication and sales authorization for all three system roles.

## 7. Sales and Financial Impact

Normal sales remain available to Admin, Pharmacist and Pharmacy Staff. The discount chosen in the Sales page dropdown affects the item discount amount and subtotal. The backend must independently reproduce the calculation before saving the sale.

For each sale item, Gross Total = verified database selling price x quantity. Discount Amount = rounded (Gross Total x selected discount percentage / 100). Subtotal = Gross Total - Discount Amount. The Final Payable Amount is the sum of item subtotals.

The sale_items record preserves the selling price, discount percentage, discount amount and subtotal used at the time of sale. Reporting and receipt templates may need to distinguish gross revenue, discounts granted and net income.

## 8. Performance and Deployment Impact

Discount-option retrieval, role checks and two-decimal monetary calculations are expected to add low computational overhead. Database reads and transactional updates should use the existing sales flow efficiently. The database, backend routes and frontend screens should be deployed together, followed by validation and regression testing in the target environment.

## 9. Testing Impact

- Verify that an Admin can add a valid discount option and that duplicate and out-of-range percentages are rejected.
- Verify that a Pharmacist cannot add, edit or deactivate an option through the Admin API and receives 403 Forbidden.
- Verify that Pharmacy Staff cannot add, edit or deactivate an option through the Admin API and receives 403 Forbidden.
- Verify that all three roles can retrieve active options and that Pharmacists and Pharmacy Staff can apply an option only through the Sales page dropdown.
- Verify that inactive options disappear from the Sales page dropdown but remain available in historical sale records.
- Verify the default 0% option and live calculations for single- and multiple-quantity items.
- Submit an unapproved, inactive, negative or out-of-range discount directly to the API and verify rejection.
- Tamper with client prices or totals; verify server calculations, sale_items snapshots and normal sales, stock, receipt and report behaviour.

## 10. Risk Analysis

| Risk | Impact | Mitigation |
|---|---|---|
| Non-admin user changes approved discount options | High | Enforce Admin role checks on every discount-management endpoint and test both non-admin roles. |

| Risk | Impact | Mitigation |
|---|---|---|
| Unapproved or inactive discount is accepted | High | Validate every selected percentage against active database options on the backend. |
| Client tampers with medicine price or sale total | High | Fetch catalog prices from the database and recalculate all monetary values on the server. |
| Incorrect historical discount records | Medium | Persist item-level price and discount snapshots in sale_items. |
| Rounding differences in sale totals | Medium | Use consistent fixed-point or decimal calculations and round to two decimal places. |
| Valid sales are disrupted by new validation | Medium | Regression-test normal sales for Admin, Pharmacist and Pharmacy Staff. |
| Inactive option remains selectable | Medium | Filter the dropdown by active status and validate status again during sale submission. |

## 11. Overall Impact and Recommendation

**Overall Impact:** Moderate - High

CR-003 is recommended because it reduces unauthorized discount use, protects medicine pricing and improves the auditability of sales. The implementation should proceed with strict role enforcement: only Admin can add or manage discount options, while Pharmacists and Pharmacy Staff can only select an existing active option from the Sales page dropdown. Backend price lookup, discount validation, database changes and regression testing for all three roles should be completed before release.